Lab 18.2 – Summarization Seq2Seq using Hugging Face

Install Libraries

```
!pip uninstall -y transformers
!pip install transformers==4.41.2 sentencepiece accelerate -q

import os
os.kill(os.getpid(), 9)
```

```
Found existing installation: transformers 4.41.2
Uninstalling transformers-4.41.2:
  Successfully uninstalled transformers-4.41.2
```

```
import transformers
from transformers import pipeline
print(transformers.__version__)
```

```
4.41.2
```

Load Summarization Model

```
summarizer = pipeline("summarization", model="t5-small")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: Use
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access pu
  warnings.warn(
model.safetensors: 100%  242M/242M [00:02<00:00, 1.21GB/s]
generation_config.json: 100%  147/147 [00:00<00:00, 13.2kB/s]
tokenizer_config.json:  2.32k/? [00:00<00:00, 169kB/s]
spiece.model: 100%  792k/792k [00:00<00:00, 1.42MB/s]
tokenizer.json:  1.39M/? [00:00<00:00, 2.41MB/s]
```

Prepare datasets

```
"""
STEP 3: Prepare Input Texts (150–400 words)
"""
```

```
texts = [

"""Artificial Intelligence (AI) is transforming industries across the globe.
It enables machines to learn from data, identify patterns, and make decisions
with minimal human intervention. AI is widely used in healthcare for diagnosi
in finance for fraud detection, and in transportation for autonomous vehicles
However, it also raises concerns about job displacement and ethical issues.
Governments and organizations are working together to ensure that AI is used
responsibly and benefits society as a whole. The future of AI holds great
potential for innovation and economic growth.""",

"""Climate change is one of the most pressing challenges facing humanity toda
Rising global temperatures, melting glaciers, and increasing sea levels are
clear indicators of environmental imbalance. Human activities such as
deforestation, industrial emissions, and excessive use of fossil fuels are
major contributors. Immediate action is required to reduce carbon emissions
and promote sustainable practices. Renewable energy sources like solar and
wind power are being adopted globally to combat climate change. Collective
efforts are necessary to protect the planet for future generations.""",

"""Machine learning is a subset of artificial intelligence that focuses on
developing algorithms capable of learning from data. It is widely used in
applications such as recommendation systems, image recognition, and natural
language processing. The effectiveness of machine learning models depends on
the quality and quantity of data available. With advancements in computing
power and data availability, machine learning continues to evolve rapidly.
However, challenges such as overfitting and bias must be addressed to ensure
accurate predictions.""",

"""Social media platforms have become an integral part of modern communicatic
They allow individuals to share information, express opinions, and connect
with others globally. While social media provides numerous benefits, it also
poses risks such as misinformation, privacy concerns, and cyberbullying.
Governments and organizations are implementing policies to regulate online
content and ensure user safety. Responsible use of social media is essential
to maximize its benefits and minimize its negative impacts."""
]

len(texts)
```

```
4
```

Generate Summaries

```
"""
STEP 4: Generate Summaries
"""

results = []

for text in texts:

    summary = summarizer(
        text,
```

```
        max_length=60,
        min_length=25,
        do_sample=False
    )[0]['summary_text']

    print("\nORIGINAL TEXT:\n", text)
    print("\nSUMMARY:\n", summary)

    results.append({
        "Original": text,
        "Summary": summary
    })
```

```
ORIGINAL TEXT:
 Artificial Intelligence (AI) is transforming industries across the globe.
It enables machines to learn from data, identify patterns, and make decisions
with minimal human intervention. AI is widely used in healthcare for diagnosis
in finance for fraud detection, and in transportation for autonomous vehicles.
However, it also raises concerns about job displacement and ethical issues.
Governments and organizations are working together to ensure that AI is used
responsibly and benefits society as a whole. The future of AI holds great
potential for innovation and economic growth.

SUMMARY:
 AI enables machines to learn from data, identify patterns, and make decisions

ORIGINAL TEXT:
 Climate change is one of the most pressing challenges facing humanity today.
Rising global temperatures, melting glaciers, and increasing sea levels are
clear indicators of environmental imbalance. Human activities such as
deforestation, industrial emissions, and excessive use of fossil fuels are
major contributors. Immediate action is required to reduce carbon emissions
and promote sustainable practices. Renewable energy sources like solar and
wind power are being adopted globally to combat climate change. Collective
efforts are necessary to protect the planet for future generations.

SUMMARY:
 climate change is one of the most pressing challenges facing humanity today .

ORIGINAL TEXT:
 Machine learning is a subset of artificial intelligence that focuses on
developing algorithms capable of learning from data. It is widely used in
applications such as recommendation systems, image recognition, and natural
language processing. The effectiveness of machine learning models depends on
the quality and quantity of data available. With advancements in computing
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However, challenges such as overfitting and bias must be addressed to ensure
accurate predictions.

SUMMARY:
 machine learning is a subset of artificial intelligence that focuses on devel

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 Social media platforms have become an integral part of modern communication.
They allow individuals to share information, express opinions, and connect
with others globally. While social media provides numerous benefits, it also
poses risks such as misinformation, privacy concerns, and cyberbullying.
Governments and organizations are implementing policies to regulate online
content and ensure user safety. Responsible use of social media is essential
```

```
to maximize its benefits and minimize its negative impacts.

SUMMARY:
 social media is an integral part of modern communication . it allows individu
```

Store Results in Table

```
import pandas as pd

"""
STEP 4: Store Results in DataFrame
"""

df = pd.DataFrame(results)
df
```

	Original	Summary
0	Artificial Intelligence (AI) is transforming i...	AI enables machines to learn from data, identi...
1	Climate change is one of the most pressing cha...	climate change is one of the most pressing cha...
2	Machine learning is a subset of artificial int...	machine learning is a subset of artificial int...

Next steps: Generate code with df New interactive sheet

Control Summary Length

```
"""
STEP 5: Compare Short vs Long Summaries
"""

text = texts[0]

short_summary = summarizer(text, max_length=30, min_length=10, do_sample=False
medium_summary = summarizer(text, max_length=60, min_length=25, do_sample=Fals
long_summary = summarizer(text, max_length=100, min_length=40, do_sample=False

print("SHORT SUMMARY:\n", short_summary)
print("\nMEDIUM SUMMARY:\n", medium_summary)
print("\nLONG SUMMARY:\n", long_summary)
```

```
SHORT SUMMARY:
 AI enables machines to learn from data, identify patterns, and make decisions

MEDIUM SUMMARY:
 AI enables machines to learn from data, identify patterns, and make decisions

LONG SUMMARY:
 AI enables machines to learn from data, identify patterns, and make decisions
```

Test Very Long Text

```
"""
STEP 7: Test with Very Long Text
"""

long_text = texts[0] + texts[1]  # combine to make longer text

summary = summarizer(long_text, max_length=80, min_length=30, do_sample=False

print("LONG TEXT SUMMARY:\n", summary)
```

```
LONG TEXT SUMMARY:
 AI enables machines to learn from data, identify patterns, and make decisions
```

Basic Analysis

```
"""
STEP 6: Basic Analysis
"""

print("""
Observations:
- The model generates concise summaries capturing main ideas.
- Short summaries may miss important details.
- Longer summaries retain more information but are less concise.
- Technical texts are summarized accurately but may simplify terminology.
- Very long inputs may lose some important details.
- Transformer models perform better than traditional extractive methods.
""")
```

```
Observations:
- The model generates concise summaries capturing main ideas.
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```

Summaries

This assignment uses pretrained sequence-to-sequence transformer models like BART to generate abstractive summaries from long text using Hugging Face pipelines